Lecture 3

Polynomial Interpolation

* More accuracy এর জন্য basis বাড়ায় দেওয়া

Degree ↑ Basis ↑
Accuracy ↑ Error ↓

$x^2 + 5x + 1 = 0$

$2x + 2x^2 = 1$

$$P_n(x) = a_0 x^0 + a_1 x^1 + a_2 x^2 + a_3 x^3 + \dots + a_n x^n$$

Degree

$$P_2(x) = a_0 x^0 + a_1 x^1 + a_2 x^2$$

Degree

Coefficient: $\{a_0, a_1, a_2\}$

Basis/Dimension: $\{x^0, x^1, x^2\} = \{1, x, x^2\}$ [3 dimension]

Degree = 2
Coeff = 3
Dim/Basis = 3
Term

Degree = Coeff - 1

n degree ⟶ n+1 basis

Function: Infinite (vector)

$f(x) = P_n(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots + a_n x^n$ [Actual function]

Tranctation error

Polynomial : Finite ⟶ $P_2(x) = a_0 + a_1 x + a_2 x^2$

* $P_n(x_i) = f(x_i)$ at a finite set of x_i (nodes)
* Polynomial can be thought as vectors in vector space (add vectors, multiply by scalar) and satisfy 10 axioms)

$5(1+x+2x^2)$

$n \uparrow$
- Accuracy বাড়বে

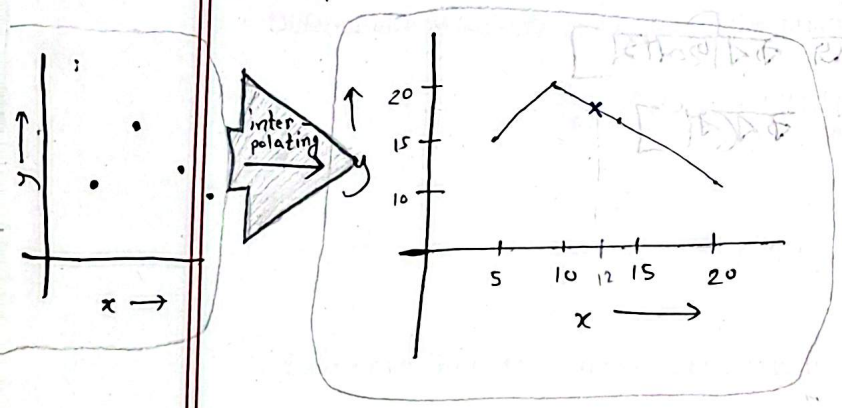
① $f(x) \in V^{\infty}$ [Infinite basis]
Belongs to ~~vector~~ set of infinite dimension

② $P_n(x) \in V^{n+1}$
Belongs to vector set of n+1 dimension

Weierstrass approximate Theorem

$P_2(x) = 2 + 3x + 4x^2 + 6x^3$
Error

Interpolation



x = 5, 10, 15, 20 এগুলো actual value জানি। এটা দিয়ে curve বানালাম। তারপর এটা দিয়ে value predict করা যাবে। এটাই interpolation.

$f(x) = a_0 + a_1 x^1 + a_2 x^2 + \ldots + a_n x^n$

If $f(x) = 1 + 2x + 3x^2 + 4x^3$ then $a_2 = 3$

Taylor Series: ← এটা দিয়ে যেকোন formula বের করা যাবে

Upto this part is degree 3 →

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)(x-x_0)^2}{2!} + \frac{f'''(x_0)(x-x_0)^3}{3!} + \ldots + \frac{f^n(x_0)(x-x_0)^n}{n!}$$

$x = 8$ [Fixed value নিয়ে কাজ করছিলাম]

$x = x - x_0$ [Range নিয়ে কাজ করবো]

Truncation error:

$$f(x) = a_0 + a_1 x^1 + a_2 x^2 + a_3 x^3 + \ldots + a_n x^n$$

$$f(x) = \sum_{K=0}^{n} \frac{f^K(x_0)}{K!}(x - x_0)^K + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}$$

(ξ → Size (Range))

Calculated portion / 1st (n+1) terms in Taylor series / Taylor polynomial of degree n.

Taylor series error / ~~Truncation~~ Lagrange Form of the remainder

Truncation error:

$$\text{Truncation error} = \left| f(x) - P_n(x) \right|.$$

As $n \uparrow$, the error reduces.

Weierstrass Approximation Theorem: as the polynomial order n increases, the approximation error reduces.

2 Taylor Series

Proof করার জন্য polynomial formula'র 'a' কে 'f' দিয়ে replace করবো

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)(x - x_0)^2}{2!} + \frac{f'''(x_0)(x - x_0)^3}{3!} + \cdots$$

Proof

Assume f can be written as a power series about x_0:

$$f(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + a_3(x - x_0)^3 + \cdots$$

→ x replaced with $x - x_0$ in function

$f(x) = a_0 + a_1 x + a_2 x^2 + \ldots$

Differentiate repeatedly (a constant differentiates to 0):

$$\frac{d}{dx}: \quad f'(x) = a_1 + 2a_2(x - x_0) + 3a_3(x - x_0)^2 + \cdots$$

$$\frac{d}{dx}: \quad f''(x) = 2a_2 + 3 \cdot 2\, a_3(x - x_0) + \cdots$$

$$f'''(x) = 3 \cdot 2\, a_3 + \cdots$$

Side note (how the power rule works):

$$\frac{d}{dx}\left(5x^2 + 2\right) = 5 \cdot 2\, x^1 + 0.$$

Now evaluate at $x = x_0$:

$$\begin{aligned}
f(x_0) &= a_0, \\
f'(x_0) &= a_1, \\
f''(x_0) = 2a_2 \quad &\Longrightarrow \quad a_2 = \frac{f''(x_0)}{2!}, \\
f'''(x_0) = 3 \cdot 2\, a_3 \quad &\Longrightarrow \quad a_3 = \frac{f'''(x_0)}{3!}, \\
&\qquad\quad a_4 = \frac{f''''(x_0)}{4!}.
\end{aligned}$$

Putting $a_0, a_1, a_2, \ldots$ back into $f(x)$, we get the Taylor series.

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3 Example: Taylor Series of $\sin(x)$sin(x)

Take $f(x) = \sin(x)$ with $x_0 = 0$. The Taylor series is

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)(x - x_0)^2}{2} + \frac{f'''(x_0)(x - x_0)^3}{3!} + \cdots$$

With $x_0 = 0$ and using $\sin(0) = 0$, $\cos(0) = 1$, $-\sin(0) = 0$, $-\cos(0) = -1$:

$$\sin(x) = \underbrace{\sin(0)}_{0} + \underbrace{\cos(0)}_{1}(x - 0) + \underbrace{(-\sin(0))}_{0}\frac{(x - 0)^2}{2!} + \underbrace{(-\cos(0))}_{-1}\frac{(x - 0)^3}{3!} + \cdots$$

$$= 0 + x - \frac{x^3}{3!} + \cdots$$

General form with error bound

Splitting into an n-degree polynomial plus a remainder (the start of the *error bound*):

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \underbrace{\frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}}_{\text{T.S. error (remainder)}}$$

where $\xi \in [0, 0.5]$.

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*All calculation in [radian]

Proof using T.S $\sin(x) = x - \frac{x^3}{3!} + \ldots$

Ans:

From TS,

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)(x-x_0)^2}{2!} + \frac{f'''(x_0)(x-x_0)^3}{3!} + \ldots$$

$f(x) = \sin x$

$f'(x) = \cos x$

$f''(x) = -\sin x$

$f'''(x) = -\cos x$

Let $x_0 = 0$

~~$\sin x = \sin(x_0) + \cos(x)(x)^2$ / 2!~~

$$\sin x = \sin(x_0) + \cos(x_0)(x-x_0) + \frac{-\sin(x_0)(x-x_0)^2}{2!} + \frac{-\cos(x_0)(x-x_0)^3}{3!} + \ldots$$

$$= 0 + x - 0 - \frac{x^3}{3!} + \ldots$$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots \quad [\text{proved}]$$

Prove $x\cos x$ using TS

$f(x) = x\cos x$

$f'(x) = \cos x - x\sin x$

$f''(x) = -\sin x - (\sin x + x\cos x)$

$= -2\sin x - x\cos x$

$f'''(x) = -2\cos x + \cos x - x\sin x$

$= -\cos x - x\sin x$

n = 2 ; find TS error of $\sin x$

↓ Degree

$\varepsilon \in [x, x_0]$

$\varepsilon \in [-0.5, 0.5]$

Ans:

$$\left| \frac{f'''(\varepsilon)\,(x-x_0)^3}{(2+1)!} \right|$$

$f(\varepsilon) = \sin\varepsilon$

$f'(\varepsilon) = \cos\varepsilon$

$f''(\varepsilon) = -\sin\varepsilon$

$f'''(\varepsilon) = -\cos\varepsilon$

$$= \left| -\cos(\varepsilon) \right| \left| \frac{(-0.5-0.5)^3}{3!} \right|$$

[$\varepsilon \in [-0.5, 0.5]$ এর মধ্যে cos value max যা পাওয়া যায়]

$\cos(0.5)$

$$= \left| -\cos(0) \right| \left| \frac{-1}{3!} \right|$$

যেটা সবচেয়ে বড় ওইটা নিবো।

$$= 1 \cdot \frac{1}{3!}$$

→ Check range end (0.5)
range start (-0.5)
Middle (0, 1)

Range 0.1 থেকে 0.5 ও দিতে পারে। তখন middle check হবে না।